

APH101 COURSEWORK - AY2324

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Please upload your report in either Microsoft Word or PDF format. If you choose to use Word, please utilize the Insert Equation feature for any mathematical symbols or equations. It is recommended to keep your report concise, as giving unnecessary details will not enhance your score and may even have a negative impact.

STATISTICAL ANALYSIS ON FACTORS INFLUENCING LIFE EXPECTANCY

Numerous studies have explored the determinants of life expectancy, focusing on demographic variables, income composition, and mortality rates. However, previous research has overlooked the influence of immunization and the human development index. This coursework seeks to develop regression models that incorporate data from 2000 to 2015 for all countries, including immunization factors (e.g., Hepatitis B, Polio, and Diphtheria), mortality factors, economic indicators, social determinants, and other health-related variables. The information on life expectancy and health factors were originally collected from the World Health Organization (WHO), while economic data was sourced from the United Nations website.

In this coursework, you need to develop a **linear regression** model for Life Expectancy and a **logistic regression** model for whether Life Expectancy is greater than 65. After building the models, you will address the following questions:

- What are the key variables influencing the life expectancy?
- What strategies should a country with lower life expectancy value (≤ 65) implement to enhance its average lifespan?

Please note:

- While a significant portion of data cleaning has already been done, there are still missing values and erroneous records present in the dataset. **It is your turn to perform further cleaning.** (*Hint*: For missing data, you may consider imputing the mean or median value from the same country to fill in the missing values. In cases where an entire column for a country contains missing values, you can use the median values from other countries to fill in the missing data. It's also important to carefully review the meaning of each column and identify potential incorrect records.)
- There is no need to include **Country** and **Year** in regression models.
- Exploratory data analysis (EDA) is required before building the model, and diagnosis of multiple linear regression is required after fitting the model. (*Hint*: If you think diagnosis results indicate the assumptions are violated, just state your reasons. Analysis can still be carried out based on the obtained model, because minor deviations from the normal distribution will not have a great impact on the fitting. *shapiro.test* is very sensitive and often rejects the null hypothesis when the amount of data is large, even if the deviation is minor.)
- You need to show and describe your process to arrive at the final model. (i.e., Do variables which have been chosen initially all affect the Life expectancy?)
- There is no standard answer to the final model. As long as your statistical model is reasonable, you will not lose points for it.
- Please elaborate on your interpretation of the model rather than just answering the question in one or two sentences.

- You need to briefly introduce background and aim at the beginning. Figures are required, while tables are optional.
- All codes used to generate all analyzes and regressions should be included as an **Appendix** at the end of the report (only R codes are acceptable).
- If you and your classmates discuss this coursework together, please thank each other by adding an **Acknowledgment** section at the end of the report (before **Appendix**). But even then, your code shouldn't be exactly the same (variable naming and coding habits often vary from person to person). If there are highly similar reports and codes, it will be suspected of cheating

COLUMN DESCRIPTION

- Status: Developed or Developing status
- Life Expectancy: Life Expectancy in age
- Adult Mortality: Adult Mortality Rates of both sexes (probability of dying between 15 and 60 years per 1000 population)
- Infant Deaths: Number of Infant Deaths per 1000 population
- Alcohol: Alcohol, recorded per capita (15+) consumption (in litres of pure alcohol)
- Percentage Expenditure: Expenditure on health as a percentage of Gross Domestic Product per capita(%)
- Hepatitis B: Hepatitis B (HepB) immunization coverage among 1-year-olds (%)
- Measles: Measles - number of reported cases per 1000 population
- BMI: Average Body Mass Index of entire population
- Under-Five Deaths: Number of under-five deaths per 1000 population
- Polio: Polio (Pol3) immunization coverage among 1-year-olds (%)
- Total Expenditure: General government expenditure on health as a percentage of total government expenditure (%)
- Diphtheria: Diphtheria tetanus toxoid and pertussis (DTP3) immunization coverage among 1-year-olds (%)
- HIV/AIDS: Deaths per 1 000 live births HIV/AIDS (0-4 years)
- Thinness 1-19 Years: Prevalence of thinness among children and adolescents for Age 10 to 19 (%)
- Thinness 5-9 Years: Prevalence of thinness among children for Age 5 to 9(%)
- Income Composition of Resources: Human Development Index in terms of income composition of resources (index ranging from 0 to 1)
- Schooling: Number of years of Schooling (years)

MARKING CRITERIA

	Statistics	Interpretation	Structure	Tables and Figures
Total Percentage	40	20	20	20
90 - 100		The best level of undergraduate coursework		
70 - 90	Statistical analysis and modeling are sufficient and logically coherent	Excellent interpretation analysis with only a small amount of room for improvement	The structure of the report is well designed and logically smooth	Tables and/or figures are well designed and have at most one or two flaws
40 - 70	The necessary statistical analysis and modeling has been completed, but there may be one or two shortcomings	Interpretation is provided, but there is still a lot to dig in	The report has a complete structure and is highly readable	Tables and/or figures convey information clearly, but there is room for improvement in the presentation
<40	Statistical analysis and modeling are imcompleted	Report lacks interpretation	Report structure is incomplete or confusing	Tables and/or figures are missing or have obvious errors